

Instructions for running the software examples included in Didactic-SoC

Running the simulation in Questasim:

- Load the Questasim environment module with "module load mentor/questasim/2023.4"
- As there is some binary incompatibility with Ubuntu 24.04 you need to run the simulation in a container. You can find the container-image and a launch script in the directory "/nas/ei/share/tools/apptainer/MSMCD"
- Launch the container and then run "make run_sim" in "\${HOME}/Edu4Chip/Didactic-SoC/sim"

Prototyping on FPGA:

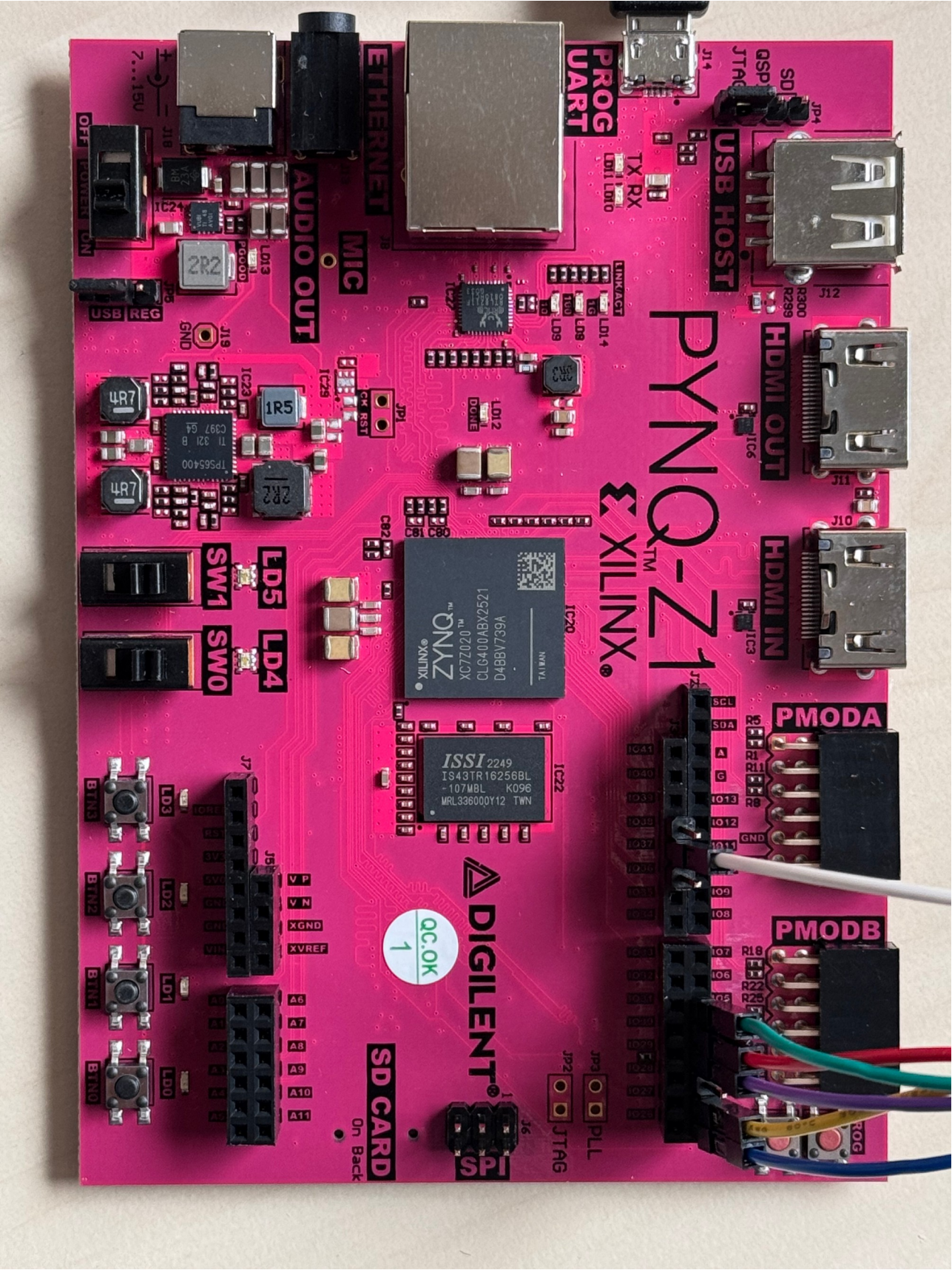
- open a new terminal and load the vivado environment module "module load xilinx/vivado/2024.1"
- open the project: "\${HOME}/Edu4Chip/Didactic-SoC/build/fpga/z1/didactic-z1.xpr"
- write the bitstream to FPGA
- connect the JTAG IO-Pins on the FPGA board to the FT4232H Module
- open a connection with openocd in a new terminal with "openocd -f \${HOME}/Edu4Chip/Didactic-SoC/fpga/utils/openocd-didactic.cfg"
- load the riscv-compiler environment modules with "module load eda_freeware/riscv/64-elf-ubuntu-24.04-gcc/2026.04.05"
- run "make load_elf" in \${HOME}/Edu4Chip/Didactic-SoC/fpga
- to start the execution: type "c" and press "Enter"
- to halt the execution: press "Ctrl+C"
- to quit the debugging session: type "quit"

To change the program that is loaded onto the IBEX-Core:

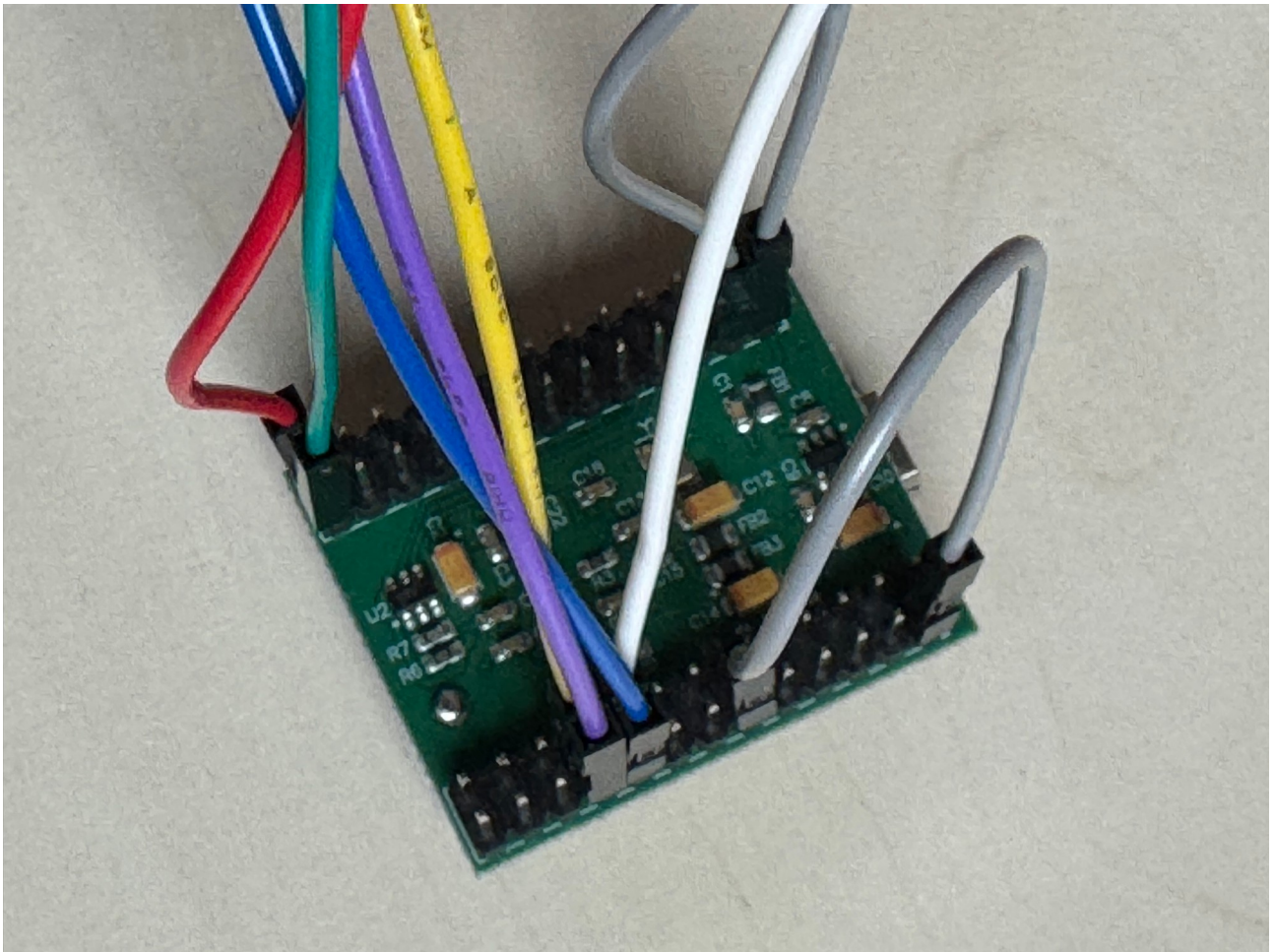
- in Didactic-SoC/fpga/Makefile or Didactic-SoC/sim/Makefile respectively change the "TEST" or "TESTCASE" variable, e.g. "TEST ?= blinky" -> "TEST ?= hello"

Wiring on the FPGA board and debugger modules:

FPGA:



FT4232H:



FT2232H:

